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Effects of pipeline distance, injectivity and capacity on CO_2 pipeline and storage site selection

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ABSTRACT

Optimising CO_2 transport and injection is a challenging issue in Carbon Capture and Storage (CCS), not only because of the complexity of the problem, but also because of timing effects when introducing new sources and/or sinks into the CO_2 transport infrastructure. In particular, the effects of storage capacity, injection site location and reservoir properties can propagate to capture and transport costs, affecting the design of the CO_2 pipelines. For example, if an injection site does not have enough capacity to store the total amount of CO_2 from a capture project, decision makers would need to consider whether to use a larger capacity site, or use the site with small capacity and later switch to a larger capacity site or use both sites.

This paper considers the effects of storage capacity, injectivity and distance to source of two sinks on optimal CO_2 transport infrastructure design and a static supply of CO_2. Optimal pipeline configurations and sink selection were determined under different combinations of CO_2 flow rate, pipeline length and storage site properties. In one scenario, two sinks both have infinite capacity but different injectivities and distances to the emission source. In the other scenario, one sink is relatively small but has a better injectivity or proximity to the emission source.

A decision tree approach was developed to provide a quick method for high-level sink selection and pipeline routing for the two scenarios based on the key project parameters including sink capacity, injectivities, pipeline distances and well cost. The scenarios where the decision trees may be useful for simplifying the design of large-scale CO_2 pipeline networks have also been analysed.

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1. Introduction

Understanding the effect of the uncertainties associated with CO_2 injection on CCS economics is a challenging problem. Although CO_2 storage accounts for a relatively small part of the total CCS costs, geological uncertainties may impose high economic risk to reservoir development (Piessens et al., 2008) and detailed storage capacity assessments are required (Welkenhuysen et al., 2013). Also, differences in the capacity and injectivity of potential storage reservoirs may significantly alter the economics and design of CO_2 capture and transport systems (Whiriskey, 2014), which may increase the complexity of CCS network optimisation. The storage capacity, injection site location and reservoir properties can all affect the design of CO_2 pipelines, as well as costs. The timing effects introduced by adding new sources and/or sinks further affect the transport network design. For example, if there are multiple storage sites available, there are many trade-offs to consider in deciding whether to use a larger capacity site or use a site with small capacity and later switch to another site.

Several methodologies have been proposed in the literature for dealing with the effects of storage capacities and reservoir properties in the CCS source-sink matching problem. For example, Oladyshkin et al. (2011a,b) consider the effect of reservoir uncertainties on CCS economics using stochastic simulations but did not include transport in their analysis. These methodologies would be acceptable for the cases where emission sources and injection locations are relatively close together (Element Energy Limited, 2010). However, they may not be applicable for cases where distances

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Nomenclature

C_i	CO_2 capacity of sink i (Gt)
F	CO_2 flow rate (Mtpa)
I_i	Injectivity for sink i (Mtpa/well)
L_i	Distance between the source and sink i (km)
R	Ratio of small sink capacity to total CO_2 capacity required over 25 years, $R = C_1/25F$
T	Time required filling the small sink using maximum flow rate (years)
W	Cost per well, A$ million per well
X	Vector of design parameters of a pipeline design and sink selection case
Y	Vector of optimal solution of a pipeline design and sink selection case

between large emitters and potential sinks are more than 1500 km (European Technology Platform for Zero Emission Fossil Fuel Power Plants (ZEP), 2011; Roussanaly et al., 2014; Wiley et al., 2015a). This limitation is significant because long transport distances require large upfront investments and so optimal sink selection and recompression design become even more important.

Other methodologies include pinch analysis (Diamante et al., 2014, 2013) and mixed-integer linear programming (Lee et al., 2014; Tapia and Tan, 2014). Both techniques were developed by analogy to existing Chemical Engineering solutions for conserving networked resources, such as cooling water and steam. These approaches are applicable to the scenarios where CO_2 sources have variable flow rates and fixed operating lives, while sinks have finite injection rates and storage capacities. The mixed-integer linear programming approaches also considered energy penalties resulting from the capture as well as sink uncertainties. However, a key assumption of these models is that the sources and sinks are in the same location, whereas the geographical distribution of sources and sinks is usually an important part of source-sink matching. Therefore, they offer limited insights into the optimal set of source-sink matches (and so the optimal CO_2 pipeline network) because the cost of building and operating the pipeline is ignored. By underestimating the cost of the whole CCS project, the network designs will be sub-optimal, especially for the scenarios where long distance transport is required (Fimbres-Weihs et al., 2011).

Our previous work (Wang et al., 2014b) investigated the lowest cost pipeline configuration and injection site selection strategy under different combinations of CO_2 flow rates, pipeline lengths and storage capacities. The research concluded that the size of the small sink is the main driver of the pipeline costs in the one-source-two-sink matching problem. This paper extends our previous work in order to gain insights into the key factors that determine the choice between two sinks with different properties. It also proposes a decision tree method that provides a short-cut approach for sink selection and pipeline design. Given a set of source and sink properties, the short-cut method proposed can quickly determine whether to use one sink over another, use both sinks concurrently, or use one and then switch to the other sink. Rather than running CCS sink-sourcing matching optimisation each time, the sink selection method developed here could be to carry out high-level CCS infrastructure assessment without running computationally intensive optimisations. As will be discussed, the method could be also used to accelerate the optimisation of existing large scale, staged CO_2 pipeline network models for certain cases (Jensen et al., 2013; Kjärstad et al., 2013; Middleton and Bielicki, 2009; Morbee et al., 2012).

2. Methodology

2.1. Framing the sink selection problem

The optimisation of CO_2 transport and injection is investigated for a simple generic CCS project. This project comprises two candidate injection sites (sinks) and a static constant supply of CO_2 (source). Two scenarios are considered, each involving one source and two sinks. The scenarios are:

- Scenario 1: The two sinks have infinite capacity but different injectivities. Also, the transport distances from the CO_2 source to the sinks vary. This scenario is applicable to cases which have an abundance of potential storage capacity (Dooley et al., 2005). As shown in Fig. 1(a), there are three non-trivial outcomes available:
- Outcome 1—Using Sink 1
- Outcome 2—Using Sink 2
- Outcome 3—Using both sink at the same time by building two pipelines.

- Scenario 2: One sink has infinite capacity and the other sink has limited capacity. In addition to variable injectivities and transport distances, the size of the limited capacity sink also varies. In this scenario, the interplay between the variables determines the optimal outcome in terms of pipeline design and sink selection. This scenario is applicable to the cases where multiple candidate sinks are required to cater for large CO_2 flow rates. For example, the Australian Carbon Storage Taskforce (2009) examined the storage capacity of a number of sedimentary basins in eastern and western Australia and found that only 35% of the candidate sinks had a 90% chance of storing more than 35 Mtpa. As shown in Fig. 1(b), there are three non-trivial outcomes available:
- Outcome 1—Using the large sink directly (LARGE)
- Outcome 2—Using both sinks concurrently (BOTH)
- Outcome 3—Using the small sink first and shift to the large one later (SHIFT).

In addition to the three outcomes listed in Scenario 2, there are others available that but they are not considered in this paper. For example, (1) the sink with better properties can be used first partially and the rest of the flow rate can be accommodated using the other sink. (2) The nearest sink can be used first for the whole flow rate at the beginning of the project. A new segment of pipeline can be built to redirect CO_2 flow from the existing sink to a new sink further away once the nearest one is filled up. This option has not been considered in this paper because we assume that the distance from the source to either sink is shorter than the distance between two sinks. This is equivalent to the condition that the angle θ between the source and the two sinks L_1 and L_2 is larger than $\arccos(L_1/2L_2)$ and $\arccos(L_2/2L_1)$ as shown in Fig. 1.

It is also worthwhile to mention that combinations of the outcomes under each scenario presented above could be used, but would probably be suboptimal due to discounting. These cases are trivial and are not considered in this paper. For example: the CCS project can use both sinks from the start and then stop using of the smaller sink; or, the sink with lower injectivity could be used first and then the sink with higher injectivity could be used later.

2.2. Assumptions

To ensure the results of this paper are as generally applicable as possible, ranges of flow rate and pipeline length as well as storage site capacity and injectivities for proposed and actual CO_2 projects are investigated as listed in Table 1. Storage capacity and properties are varied because they have been proven to be a crucial factor for

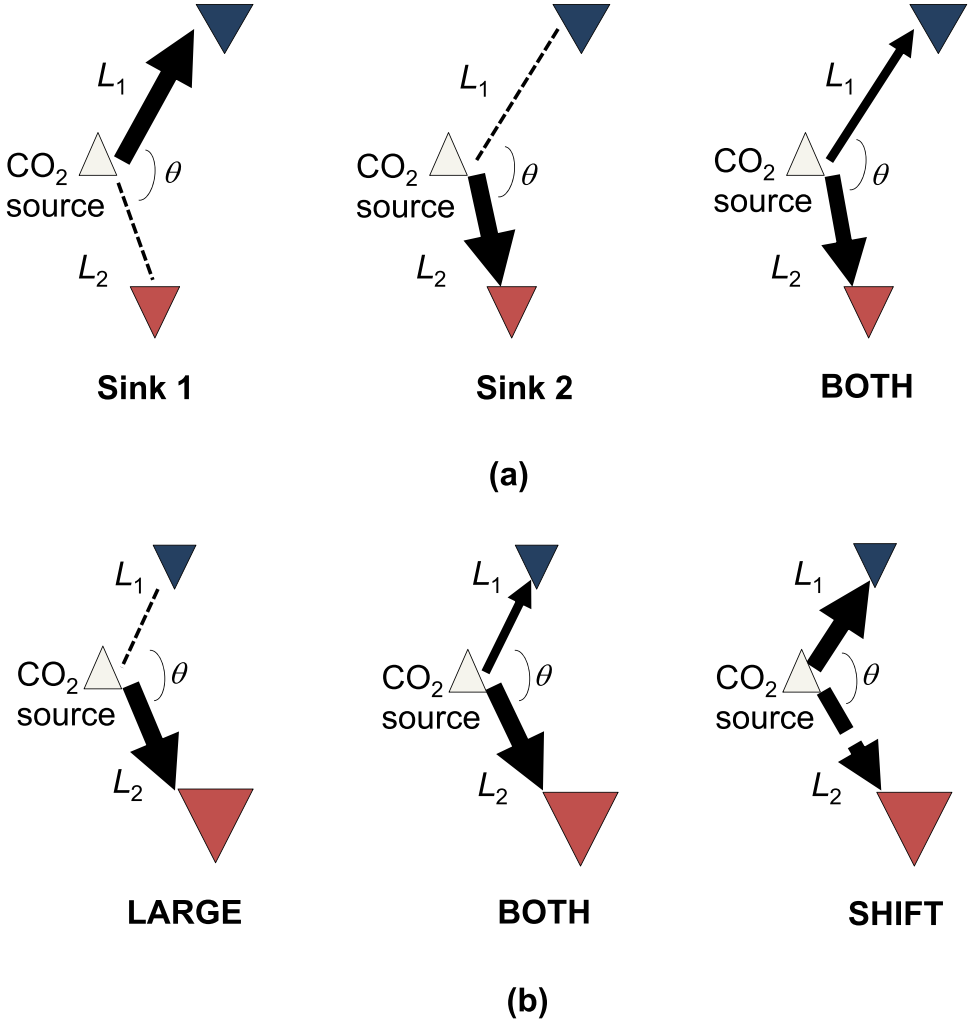


Fig. 1. Three design outcomes of two typical two-sink competing scenarios. (a) Scenario 1: each sink has infinite storage capacity. (b) Scenario 2: one sink has infinite capacity and the other sink has limited capacity.

Table 1
Ranges of the variables used in the three cases.

Properties used in simulation		Scenario 1: Two sinks have infinite capacity	Scenario 2: Only one sink has infinite capacity
F	CO_2 flow rate (Mtpa)	15, 20, 25, 30, 35	15, 20, 25, 30, 35
R	Ratio of small sink capacity to total CO_2 capacity required over 25 years	Infinite	0.1, 0.3, 0.5, 0.7, 0.9
L_1	Pipeline length from CO_2 source to sink 1 (km)	100, 300, 500	100, 300, 500
L_2/L_1	Ratio of the distance between the source and sink 2 to the distance between the source and sink 1	1.5, 3, 6	0.1, 0.2, 1, 5, 10
I_1/I_2	Ratio of the injectivity of sink 1 to that of sink 2	1/6, 1/3, 2/3	0.1, 0.2, 1, 5, 10
W	Total unit cost of drilling and completing a CO_2 injection well, A$ million per well	2, 5, 8	2, 5, 8

CCS implementation in different countries (Viebahn et al., 2014, 2012).

Fig. 2 illustrates the locations of worldwide sources and sinks. The pipeline lengths used in this paper (see Table 1) cover most of the possible cases in CCS pipeline optimisation (Wang et al., 2014a). For example, in North America a large percentage of CO_2 sources and potential sinks are within 150 km (Element Energy Limited, 2010). On the other hand, the upper bound for transport distance is applicable to many potential CCS scenarios, particularly in Europe and Australia (European Technology Platform for Zero Emission Fossil Fuel Power Plants (ZEP), 2011; Fimbres Weihs and Wiley, 2012; Morbee et al., 2012; Stewart et al., 2012; Wiley et al., 2015a).

The flow rates selected for the CO_2 transport and injection range from 15 Mtpa to 35 Mtpa. The range is indicative of the flow rate in a trunk line receiving CO_2 flow from multiple CO_2 sources in existing studies (Fimbres Weihs and Wiley, 2012; Fimbres-Weihs et al., 2011). The minimum injection rate of 15 Mtpa corresponds to a total storage capacity of 0.375 Gt over 25 years, while the maximum injection rate of 35 Mtpa corresponds to a storage capacity of 0.875 Gt over the same time frame. The number of wells is estimated based on the assumed injectivity and the CO_2 flow rate.

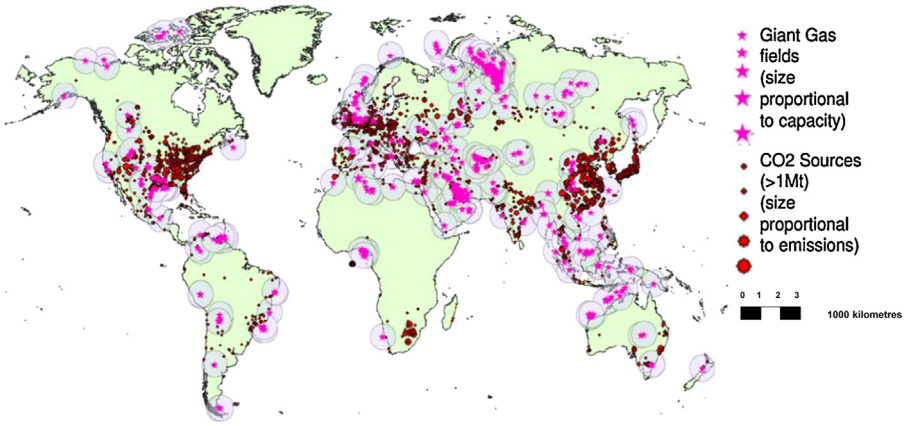


Fig. 2. Global CO_2 source (red circle) and sink (pink star) illustrated by Element Energy Limited (2010). Circles around pink stars indicate distances within 600 km of each potential sink. Red circles indicate candidate CO_2 sources larger than 1 Mtpa. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Assessing the effect of reservoir heterogeneity and well interference is outside the scope of this work.

The economics of the capture plants are not considered in this paper, since capture economics can be optimised separately from pipeline design in the CCS chain (Fimbres Weihs and Wiley, 2012).

Other assumptions used in this paper include,

- The distance between the two sinks is greater than that from either sink to the source. Therefore it is not worthwhile to extend the pipeline from one sink to the other.
- The life time of the project is 25 years.
- The cost per well is either A$2 million/well, A$5 million/well or A$8 million/well. This covers the typical range of drilling and completion costs under a wide range of reservoir properties and economic variables, such as depth, rock type and oil prices (or rig availability) (Carbon Storage Taskforce, 2009; Toews and Naumov, 2015).
- The costs of building, operating and decommissioning the CO_2 transport and injection facility are estimated in 2012 Australian dollars.
- A real discount rate of 7% is used.
- The pipeline steel grade is X65.
- We assume that the CO_2 is compressed to 8 MPa in the capture unit so that it is ready to be transported in the pipeline in a dense state at 25 °C. The outlet pressure of the transport pipeline is 8 MPa. The CO_2 in the pipeline is compressed at the start of the pipeline and recompressed using booster pump(s) as required to maintain the CO_2 in a dense state for transport along the pipeline.
- The maximum allowable working pressure is 15 MPa.

The objective function that is minimised (Eq. (1)) is the specific cost of CO_2 transported and injected.

$$f_{obj} = \frac{PV(Capex_{Pipe} + Opex_{Pipe} + Capex_{Booster} + Opex_{Booster} + Capex_{Well} + Capex_{Decommission})}{PV(CO_2\text{transported and injected} - CO_2\text{generated by transport and boosters})} \quad (1)$$

The calculations are conducted in line with standard CO2CRC methodology (Allinson et al., 2012, 2006; Wiley et al., 2015b) and performed using ICCSEM, a CCS economic optimisation model developed by UNSW Australia for the CO2CRC.

The pipelines are designed following Australian Standard AS2885.1 (Standards Australia, 2012). The thickness is determined according to commonly used industrial practice (Cosham and Eiber, 2007).

It should be noted that this paper only considers onshore pipeline transport and storage. Other transport scenarios (not considered in this paper) may involve offshore pipelines, as well as truck and/or ship transport.

- Under the offshore pipeline scenario, more complex trade-offs are involved. For example, booster pumps will be impractical offshore which means that, for the same flow rate, a larger diameter pipeline might be required. Any gravity head gain due to changes in seabed elevation may increase the allowable pressure drop and therefore reduce the pipeline diameter. Also, inlet pressure can be another key parameter for optimising offshore pipeline systems (Knoope et al., 2014).
- Under truck or ship transport scenarios, variables such as carrier capacity, service speed, fleet size and schedule would also need to be considered.

Since the primary purpose of this paper is to demonstrate a methodology, these scenarios were not considered, however it would be relatively straightforward to include them if required (Weihs et al., 2014; Wiley et al., 2015a).

The total injectivity of a storage site is usually determined by the areal extent, thickness, depth, permeability, porosity and well spacing. We assume that the injectivity per well ranges from a least preferable value of 0.3 Mtpa per well to a most preferable value of 1.5 Mtpa per well. Since this paper is targeted at high level assessment of CCS project, the storage cost (A$ million) for a specific sink is calculated in a simplified form: the CO_2 flow rate (Mtpa) is divided by the injectivity per well (Mtpa/well) and then multiplied by the well cost (A$ million/well). In a more detailed injection design, consideration would be given to the effect of well spacing, well numbers and sweet spots in the formation, as well as the option of producing formation water to manage reservoir pressure. The final choice of injection design will involve trade-offs, since each of these variables will have an economic impact on the total system cost (Michael et al., 2011; Neal et al., 2013, 2011).

For the purpose of this paper, we did not consider the effect of well interference and assume that the number of wells is linearly related to the flow rate of CO_2. If well interference is considered, the number of wells would increase with flow rate at an increasingly faster rate. This would intensify the trade-off between transport and injection and most likely reduce the chance of using the sink with poor injectivity.

2.3. Decision tree method

The problem of pipeline design and sink selection can be interpreted using a decision tree. The decision tree method is one of the most useful statistical techniques for classifying discrete finite results, by mapping observations about several parameters to a set of target values (Conca, 2014). The method can be used to visually and explicitly represent the decision making process. A categorical decision tree consists of leaf nodes representing class labels (outcomes) and branches representing conjunctions of features (set of variable values) that lead to those class labels. The tree also contains non-leaf nodes where a combination of values of input variables can be used to determine which branch should be used next. Every leaf node of the tree is labelled with an optimal design outcome. More specially, each optimal design decision, Y, can be described as a function of the set of input vector, X,

$$\begin{pmatrix} Y_1 \\ \ldots\ldots \\ Y_n \end{pmatrix} = f \begin{pmatrix} X_1 \\ \ldots\ldots \\ X_n \end{pmatrix} = \begin{pmatrix} OptimalOutcome_1 \\ \ldots\ldots \\ OptimalOutcome_n \end{pmatrix} \quad (2)$$

where X_i is a vector of design parameters for the CO_2 transport and injection case i,

$$X_i = [R_i, F_i, L_{1,i}, L_{2,i}, W_i, I_{1,i}, I_{2,i}]. \quad (3)$$

where R is the ratio of small sink capacity to total CO_2 capacity required over 25 years. F is the total flow rate. L is the pipeline length, W is cost per well and I represents the injectivity per well.

The X and Y data can be used to develop a categorical decision tree algorithm (Quinlan, 1986) which determines the nodes of the decision tree by iteratively calculating entropy $H(S)$ (defined in Eq. (4) below).

$$H(S) = -\sum_{y \in Y} p(y)\log_2 p(y) \quad (4)$$

where S is the current dataset for which the entropy is to be calculated. Y is the set of design decisions and $p(y)$ is the proportion of the number of each design decisions y to the total number of design decisions in S.

In each step, the decision tree algorithm chooses one design parameter as a node for which entropy is minimised. This parameter divides S into subsets, where the iterative method would be applied again until there is only one design decision in the subsets.

The performance of derived trees can be validated using the cases in Table 1. The accuracy is quantified as the number of correct predictions divided by the total number of predictions. Data manipulation may also be applied to simplify the decision tree. As will be discussed in the following section, we compared the results using individual variables and a reduced set of variables obtained by taking ratios.

3. Result analysis

3.1. Cost drivers

Fig. 3 illustrates the breakdown of levelised transport and total costs (respectively) for different flow rates. For transport costs, the cost of each component decreases gradually as the flow rate increases because of economies of scale (Chandel et al., 2010).

The optimal solutions to this sink selection problem under the two scenarios are largely determined by the trade-offs between the transport cost and injection cost. As flow increases, the low injection cost of the sink with larger injectivities will have a greater impact when the flow rate gets larger because economies of scale reduce the transport cost to both sinks. For example, under Scenario 1, when the low injectivity of the closer sink results in a higher cost premium than the extra cost caused by longer transport distance to the second candidate sink, the sink further away with better injectivity is more likely to be used. Fig. 4 shows the trade-off between transport and injection cost under Scenario 1 when the CO_2 flow rate is 15 Mtpa and the two sinks are 300 km and 450 km away from the source. The more distant sink (Sink 2) has better and constant injectivity (1.5 Mtpa/well). When the injectivity ratio increases, the storage cost of using the closer sink generally decreases. In other words, as the injectivity ratio increases, the closer sink with poor injectivity gradually becomes more economical until the trade-off point in the injectivity ratio (I_1/I_2) is reached somewhere between a value of 0.2 and 0.3. Similar trade-offs can be found under other combinations of flow rate, injectivity ratios, pipeline length ratios and well cost.

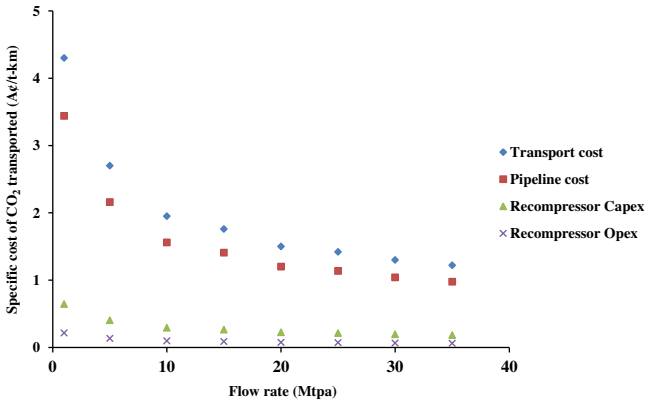


Fig. 3. Levelised cost under different flow rates when L = 250 km, the pipeline operates for 25 years.

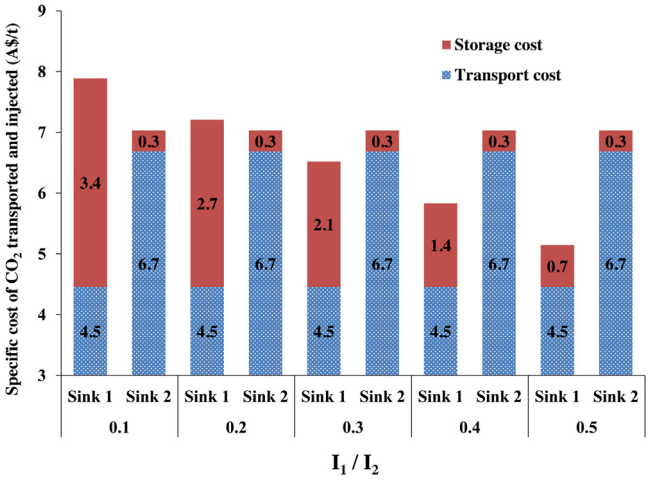


Fig. 4. Transport and injection cost breakdown under different I_1/I_2, when F = 15 Mpta, L_1 = 300 km, L_2 = 450 km, W = 6 A$ million per well, I_2 = 1.5 Mtpa/well.

3.2. Scenario 1: two infinite-capacity sinks

In Scenario 1, it should be noted that there are many pipeline design options available. For example, when two sinks are used at the same time, each sink will share a certain percentage of the total flow rate. However, it can be easily proven that it is always more economical to use one of the two sinks directly, rather than using both of them together. This is because using only one sink leads to building a large pipeline which can achieve greater economies of scale. For example, Fig. 5 shows the economics of storing different proportions of the total CO_2 flow rate in each sink. The results show

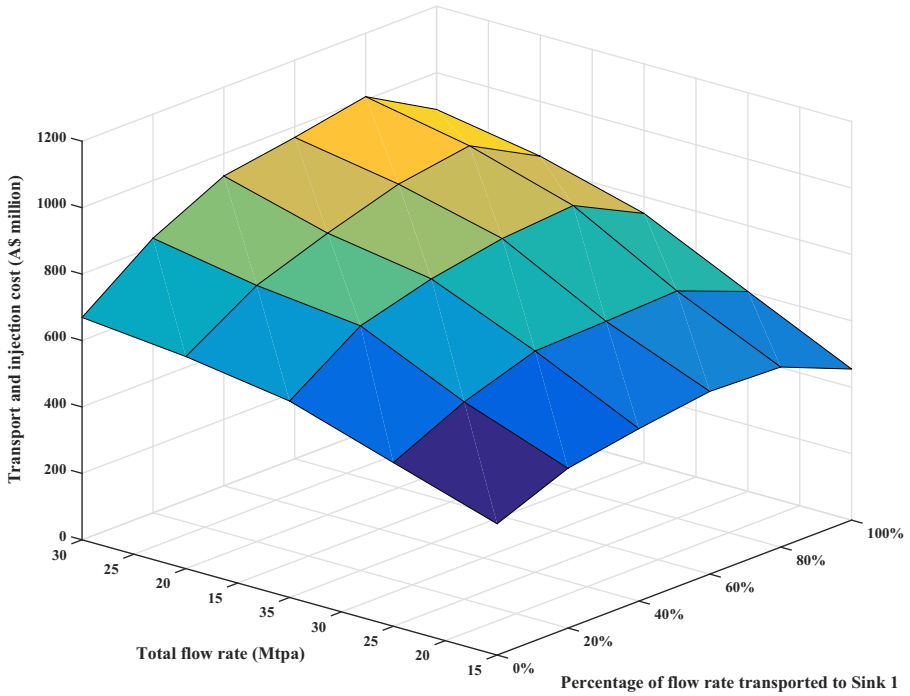


Fig. 5. Effect of F and its distribution between sink on transport and injection cost. L_1 = 100 km, L_2 = 150 km, W = 2 A$ million per well, I_1 = 0.125 Mtpa/well, I_2 = 0.75 Mtpa/well.

Table 2
Number of inputs, complexity and accuracy of decision tree in Scenario 1.

Decision tree version	Inputs	Number of Inputs	Decision tree levels	Accuracy
1	F, L_1, L_2, I_1, I_2, W	6	12	96.9%
2	L_1, L_2, I_1, I_2, W	5	10	96.7%
3	$F, L_2/L_1, I_1/I_2, W$	4	7	95.5%
4	$L_2/L_1, I_1/I_2, W$	3	3	95%

that for a given total flow rate, storing in only one sink has lower total costs than using the two sinks at the same time under Scenario 1.

Other options, such as using one sink at the beginning of the CCS project and switching to the other one later also lead to a higher CO_2 avoidance cost. This additional cost is because more pipelines and wells are installed despite the fact that both sinks have sufficient capacity for the project. Therefore, it is only worthwhile to compare the avoidance cost of using the two sinks separately.

The economics of storing different amounts of CO_2 in the two sinks with infinite capacity but varying injectivities and distances to the source were evaluated using the ICCSEM optimisation model. Decision trees were developed for each of the combinations of input variables (X) shown in Table 2. The outcome of the decision tree is the choice of sink for injection (Y) in either sink 1 or sink 2 as obtained from the optimisation.

As shown in Table 2 for Scenario 1, the variables F, L_1, L_2, I_1, I_2 and W are required to achieve an overall accuracy of 96.9% in a complex decision tree of 12 levels. Using only the variables $F, L_2/L_1, I_1/I_2$ and W results in a simpler, 3-level tree (shown in Fig. 6) with an overall accuracy of 95%. An accuracy of 95% is usually considered reasonable for decision trees under different applications (Friedl and Brodley, 1997; Tien Bui et al., 2012). Comparing accuracy, computational time and storage uncertainty, the method proposed here can help users carry out a quick screening across all the candidate pipeline design options without running computationally intensive optimisations.

3.3. Scenario 2: one of the two sinks has limited capacity

In Scenario 2, it should be noted that all three design options (LARGE, BOTH, SWITCH) could be chosen as the optimal outcome, which is consistent with our previous paper (Wang et al., 2014b).

Fig. 7 gives the cost surfaces of the three different design outcomes under a specific combination of CO_2 flow rate (F = 25 Mtpa) and small sink size (R = 0.5). The optimal design surface combining the three outcomes is shown Fig. 7(d).

For example, when injectivity ratios I_1/I_2 are less than 1, using the large sink directly (LARGE) is the optimal design. When I_1/I_2 increases to 1 and L_2/L_1 is any value between 1 and 10, using the two sinks at the same time (BOTH) has the lowest cost. The rest of minimum cost surface illustrates that using one sink with limited capacity first and then switching to the infinity capacity sink will have a lower transport and storage cost. This process can be repeated for other combinations of injectivity ratio and pipeline length ratio to derive the optimal design. Similarly, the data can be graphed using different axes to allow the determination of the optimal design for different combinations of CO_2 flow rate and small sink size.

The graphical representation in Fig. 7 has its limitations. First, it is only applicable to a certain set of values of injectivity, pipeline length and storage capacity data. Therefore a new minimum cost surface is required for every set of data. Second, it is hard to determine visually which option gives the lowest cost. This is especially true when the injectivity ratio or pipeline length ratio lies between the grids of the cost surface and requires interpolation.

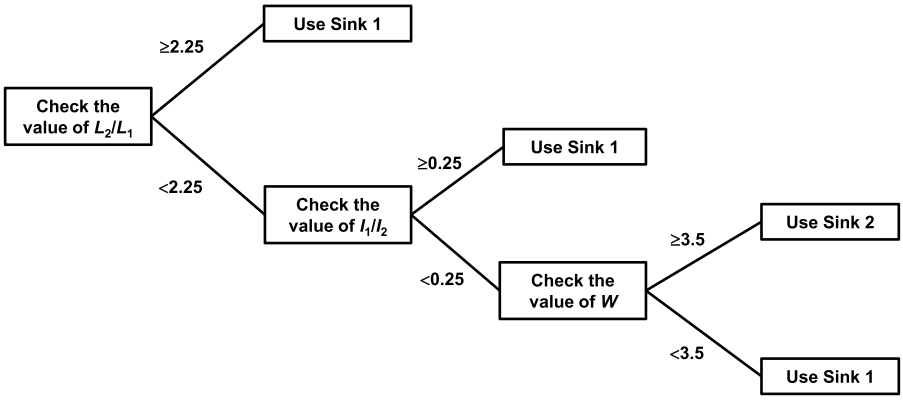


Fig. 6. Decision tree under Scenario 1 for two infinite-capacity sinks.

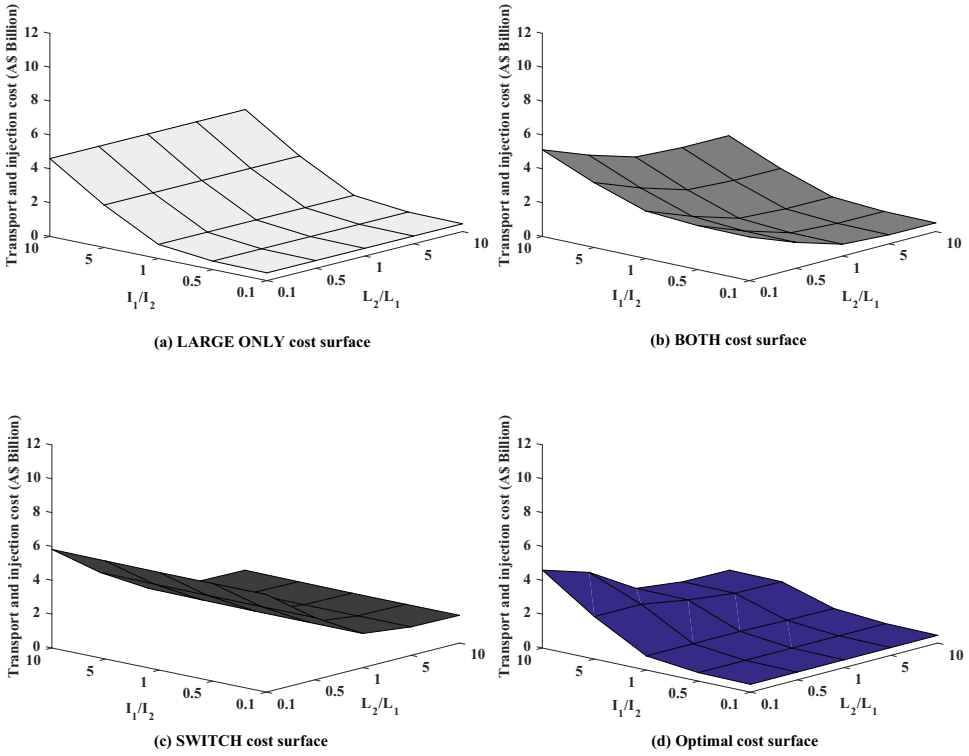


Fig. 7. Cost surfaces of three different design options and optimal cost design under a scenario where F = 25 Mtpa, R = 0.5, W = 5 A$ million per well.

Table 3
Number of inputs, complexity and accuracy of decision tree in Scenario 2.

Decision tree version	Inputs	Number of inputs	Level of decision tree	Accuracy
5	R, L_1, L_2, I_1, I_2, W	6	18	96.3%
6	R, L_1, L_2, I_1, I_2	5	13	95.3%
7	$R, L_2/L_1, I_1/I_2, W$	4	6	95.2%
8	$R, L_2/L_1, I_1/I_2$	3	5	92.2%

Conversely, the decision tree method described in this paper does not have these limitations. Applying this method to Scenario 2 with six input variables (R, L_1, L_2, I_1, I_2, W) resulted in a decision tree with an overall accuracy of 96.3% but was quite complex and involved 18 levels. A simpler tree using the variables R, L_2/L_1, I_1/I_2 and W required only 6 levels and gave an overall accuracy of 95.2%, as shown in Table 3. Comparing the graphical method shown in Fig. 7 and the decision tree in Fig. 8, it is clear that the latter is more user-friendly, covers a wider range of transport and storage scenarios and is faster to employ.

An analysis of the two scenarios showed that the absolute values of the pipeline length and the injectivities are not significant in deciding the outcome of this two-sink selection problem. Using pipeline length ratio L_2/L_1 and injectivity ratio I_1/I_2 gives satisfactory results. By reducing the number of inputs, a significantly simpler decision tree is obtained with only marginal reductions in

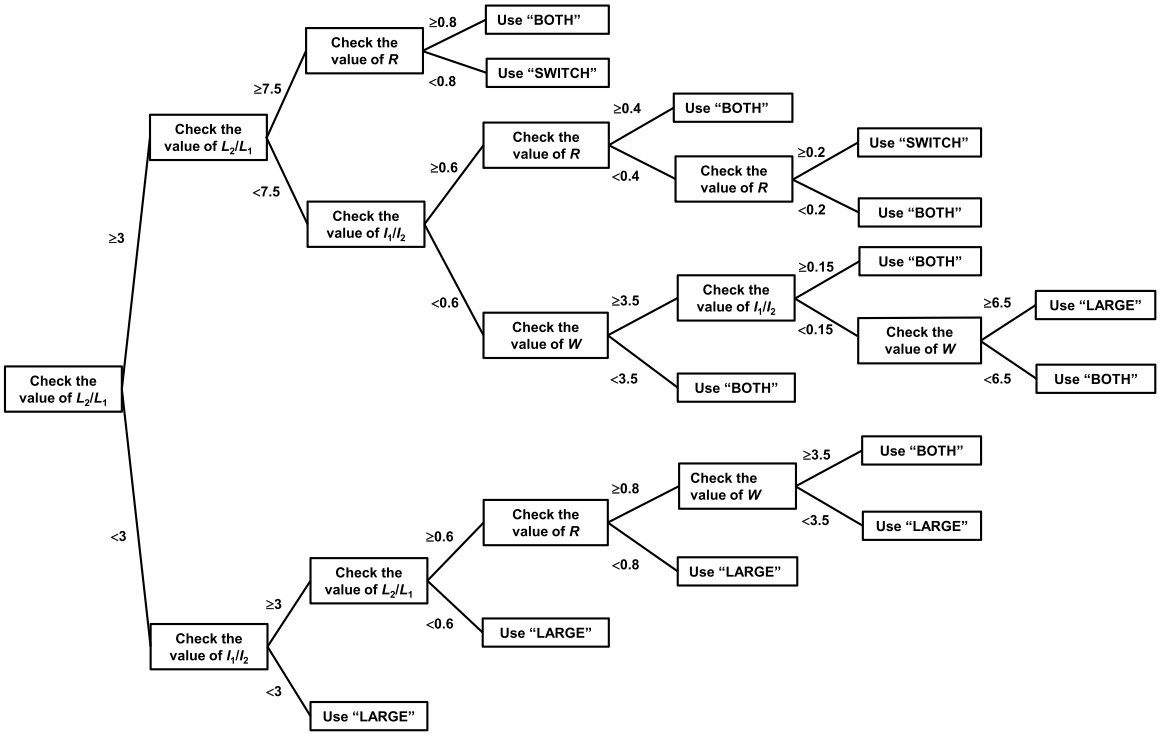


Fig. 8. Decision tree under Scenario 2 when one of the two sinks has limited capacity.

accuracy. Moreover, the results from Scenario 1 are much simpler because there are fewer inputs and outcomes involved.

Another important point to note is that the cost per well (W) could be different for the two sinks. For example, costs of transporting the drilling equipment to the two storage sites are different. However, since this research targets high-level, early stage assessment of a CCS project, differences in well costs in the two sinks are not considered. We assume the cost of drilling one well in each sink is the same. In practice, the cost per well (W) could be different for the two sinks, for example they may have different mobilisation costs or well depths. Furthermore, there are other variables that could also be included in the analysis, such as differing topography, whether sinks and pipelines are onshore or offshore, etc. Including the effects of them would add extra dimensions to the decision tree.

It should be noted that worldwide, pipeline and compression costs may vary under different scenarios. Variability in pipeline costs is largely associated with the pipeline length and uncertainty surrounding the steel price and construction costs. Further, construction costs are strongly related to labour costs and the geography of the pipeline route. The sensitivities of optimal pipeline design to variations in pipeline length are reflected in the results. The steel price and construction costs used in this paper are based on an Australian context (Allinson et al., 2012). As steel prices and construction costs vary, the pipeline capex may increase or decrease. However, changes in steel price (and to a lesser extent construction costs) are generally a function of time and vary with supply and demand. This means that, for pipelines built at the same time, the effect of variations in the steel price and therefore pipeline cost is effectively cancelled out because a key variable involved in the simplified decision tree is L_2/L_1. Similarly, labour costs are generally a function of time for a given location and any variability is again cancelled out when comparing pipeline options. The effect of geography on pipeline construction cost is very site specific and so was not included, since the purpose of the scenarios presented in this paper is to demonstrate a methodology.

As shown in Fig. 3, the recompression Capex and Opex represent around 9% of the transport cost (on average). Based on our calculation, a 30% change in recompression cost will affect the total transport cost by 4.5% at most.

4. Case studies

In this section, the decision tree methodology is applied to the two cases described in Table 4

- *Case study 1:* Both sinks have infinite capacity. Sink 1 is further away and has a higher injectivity.
- *Case study 2:* The small sink has the capacity to store 0.25 Gt which is not enough to store CO_2 for 25 years. It is closer to the CO_2 source than the large sink and has better injectivity.

Using the developed decision trees (shown in Figs. 6 and 8), the outcomes of these two cases can be easily determined without running any complex calculations. In case study 1, the choice of sink can be determined using Fig. 6. The decision process begins with the leftmost node and moves through various nodes to the final outcome. At the first node, the case's pipeline length ratio (equal to 2) is compared against 2.25 to determine which branch to follow. Then the injectivity ratio (1/5) and well cost (5) need to be checked. Finally, the decision to use Sink-2 will be made (using sink 2 only, which is further away but has a higher injectivity).

In *Case study 2*, a similar process is followed based on Fig. 8 and to arrive at the decision that the two sinks should be used at the same time (BOTH option).

Through analysis of these case studies, the proposed decision tree method, based on a combination of sink capacity, pipeline lengths and injectivities, is significantly quicker than conventional optimisation for estimating the optimal sink selection and pipeline design. The time required to arrive at the sink selection using different methods for the two case studies is given in Table 5. When compared against an existing optimisation method

Table 4
Data used in the case studies.

Values used in the case study	Case 1	Case 2
Capacity of sink 1 in Gt, C_1	infinite	0.25
Capacity of sink 2 in Gt, C_2	infinite	infinite
CO_2 flow rate from CO_2 emission cluster in Mtpa	20	20
Cost per well in A$ million, W	5	5
Ratio of small sink capacity to total CO_2 capacity required over 25 years, R	N/A	0.5
Pipeline length ratio, L_2/L_1	2	3
Injectivity ratio, I_1/I_2	1/5	3

Table 5
Comparison of computing time between the decision methods used in the two case studies.

Methods used in Case 1	Computing time in seconds Case 1	Methods used in Case 2	Computing time in seconds Case 2
Decision Tree 1	0.045	Decision Tree 5	0.065
Decision Tree 2	0.044	Decision Tree 6	0.059
Decision Tree 3	0.039	Decision Tree 7	0.056
Decision Tree 4	0.039	Decision Tree 8	0.055
ICCSEM optimisation	19.233	ICCSEM optimisation	28.325

Table 6
Scenarios where the decision tree methodology can be used to simplify source-sink(s) matching calculations.

Sink availability	Sink usage scenario			
	Use only one sink	Use two sinks concurrently	Use two sinks sequentially	Use three or more sinks concurrently or sequentially
One sink available	Not required			
Two sinks available, distance between two sinks is shorter than the distance from the source to either of the two sinks	Applicable but using a different decision tree required			
Two sinks available where the distance from the source to either sink is shorter than the distance between two sinks	Decision tree in this paper applicable			
More than two sinks available and pairwise comparison can be used				Use other optimisation methods
More than two sinks available and pairwise comparison cannot be used		Use other optimisation methods		

(Fimbres-Weihs et al., 2011; Wang et al., 2014b), decision trees give an answer in the order of 500 times quicker. Thus, the decision tree method could be used to carry out quick, high-level, early stage CCS assessment.

5. Decision tree method and large network models

The method presented in this paper may also be integrated into existing source-sink matching algorithms to reduce the computational complexity of high-level CCS transport and injection planning. More specifically, these models can usually be divided into sub-cases where there is a constant CO_2 supply and several candidate storage options. Optimising these cases usually involves comparing the costs of all the possible source-sink(s) combinations and is very computational intensive. The applicability of the decision tree method to different scenarios is summarised in Table 6. The scenarios where the decision trees proposed in this paper can simplify source-sink(s) matching calculations are:

- When there are two sinks available for CO_2 storage and the angle θ_i between the source and two sinks (as shown in Fig. 1) is larger than both $\arccos(L_1/2L_2)$ and $\arccos(L_2/2L_1)$, the decision tree methods proposed in this study can be used. Users should select the appropriate decision tree based on the sink capacities (see Sections 3.2 and 3.4). An example of this scenario can be found in the south and east of Spain (Morbee et al., 2012), where each CO_2 hub has two candidate storage sites available.
- When there are more than two sinks available for the CO_2 storage and the angle θ_i between the source and any two sink combinations is larger than $\arccos(L_1/2L_2)$ and $\arccos(L_2/2L_1)$, then pairwise comparisons can be applied to find the most favourable pipeline connection between the source and any two-sink pairs. Generic transport and storage cost models, such as Mohitpour et al. (2006) and McCollum and Ogden (2006) could then be used to calculate and rank the costs of these favourable pipeline connections. This method will save the network models from exhaustively checking the actual costs of all the candidate pipeline connections between the source and two-sink pairs. This scenario is exemplified by the West Poland case shown in Morbee et al. (2012), where the CO_2 hub may have more than two candidate storage sites available.

The scenarios where the decision tree method developed in this paper is not suitable are:

- Cases where there is only one sink available for storage. The decision tree method is not required as the transport cost can be directly calculated using generic pipeline and storage cost models.
- When the distance between two sinks is shorter than the distance from the source to either of the two sinks (i.e. the angle θ_i subtended between the source and the two sinks is less than $\arccos(L_1/2L_2)$ and/or $\arccos(L_2/2L_1)$), the decision tree developed in this paper is not applicable. However, a different decision tree could be developed for this scenario.
- When three or more sinks are required to be used concurrently or sequentially, the decision tree becomes overly complex and is not expected to offer an improvement in computing time. Algo-

rithms connecting more than three sinks, such as Kazmierczak et al. (2009), should be used to determine optimal pipeline routes. However, with further detailed analysis, it may be possible to develop other decision trees for some of these cases.

6. Conclusion

In this paper, optimal pipeline configurations are determined under different combinations of CO_2 flow rates, pipeline lengths, injectivities and storage capacities under two generic scenarios. The economics of three possible transport and storage design outcomes are discussed for two scenarios, namely (1) two sinks have infinite capacity but different injectivities and distances from the CO_2 source and (2) one sink have infinite capacity and the other sink has limited capacity.

The decision trees developed are for CCS scenarios with one CO_2 source (or a hub of CO_2 sources) and two candidate sinks with fixed parameters. To simplify the problem, the distance between the two sinks was assumed to be greater than the distance of either sink to the source.

When one of the sinks has limited capacity, the results indicate that under some circumstances there are benefits in using a sink with less capacity than the project requires. The trade-offs between the transport and injection costs have been quantified using a decision tree method. Though generic CCS cases are investigated in this paper, the data used is within the range of expected CCS scenarios and can be applied to different CCS datasets around the world. The decision trees developed can be used to make a quick and reasonably accurate estimate of optimal sink and pipeline designs without running any optimisation. Further, the two case studies conducted indicate that using decision trees can accelerate the calculation speed significantly compared to the existing optimisation method. Implementing these techniques would result in significant time and cost savings for scoping analyses. The paper also discussed the scenarios under which the decision tree methods could be integrated in multiple-source multiple-sink network optimisation. Using the decision tree method it may be possible to reduce the size of the problem space for optimisation by quickly screening a large set of possible source-sink(s) combinations.

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